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Q1. What is OFDM and how it differs from FDM?

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A1. *Multiplexing.* It is often desirable to transmit several messages over a single channel. Multiplexing is a technique which enables this. There are number of

is then transmitted along the single channel by electromagnetic means or transmission lines. FDM are commonly used in radio broadcasts, TV networks and voice transmission in telephone network.

Orthogonal Frequency-Division Multiplexing (OFDM). The basic idea of OFDM was introduced and

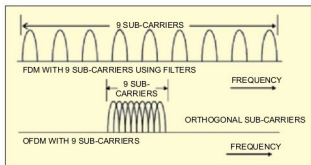


Fig. 1: Difference between bandwidth requirements in FDM vs OFDM

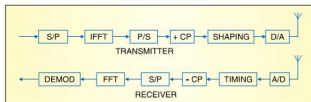


Fig. 2: OFDM transceiver block

techniques available based on time (TDM) and frequency (FDM). Frequency-division multiplexing (FDM) and orthogonal frequency division multiplexing (OFDM) are two such multiplexing techniques based on frequency.

Frequency-division multiplexing (FDM). In frequency division multiplexing, the available bandwidth of a single physical medium is subdivided into several independent frequency channels. Independent message signals are translated into different frequency bands using modulation techniques, which are combined by a linear summing circuit in the multiplexer, to form a composite signal for transmission. Each FDM channel is separated from the others by a frequency guard band to reduce interference between adjacent channels. The composite signal

multi-path interference.

The benefits of OFDM are high spectral efficiency, resiliency to RF interference and lower multi-path distortion. OFDM is used in high-speed transmission over telephone lines, such as digital sub-carrier lines, digital audio broadcasting and digital cellular communication systems.

The digital base band of an OFDM transmitter consists of six basic blocks, as shown in Fig. 2. Available spectrum is divided into 'n' orthogonal sub-carriers. The encoded data is mapped onto the sub-carriers using an n-point IFFT. Finally, an n-sample long cyclic prefix (+CP) is added to reduce the inter symbol interference (ISI). Block diagram of an OFDM transceiver is shown in Fig. 2.

Q2. What does the arrow mark in a transistor circuit denote?